



Physics HL/SL A.1 Kinematics	Physics HL/SL A.1 Kinematics	Physics HL/SL A.1 Kinematics
Why can horizontal and vertical motion be analysed independently in projectile motion?	A ball is thrown straight up. What is its acceleration at the very top of its path?	An object falls from rest for 3.0 s. How far does it fall ($g = 9.81 \text{ m}\cdot\text{s}^{-2}$)? A 15 m B 29 m C 44 m D 88 m
IB-PHYSICS-A.1-1 CONCEPT	IB-PHYSICS-A.1-2 CONCEPT	IB-PHYSICS-A.1-3 MULTIPLE CHOICE
Physics HL/SL A.1 Kinematics	Physics HL/SL A.2 Forces and momentum	Physics HL/SL A.2 Forces and momentum
A cyclist decelerates uniformly from $8.0 \text{ m}\cdot\text{s}^{-1}$ to rest over 20 m. Find the deceleration.	State Newton's third law and explain why the two forces in a pair do not cancel each other out on the same object.	Why is momentum conserved in a collision but kinetic energy generally is not?
IB-PHYSICS-A.1-4 PRACTICE	IB-PHYSICS-A.2-1 CONCEPT	IB-PHYSICS-A.2-2 CONCEPT
Physics HL/SL A.2 Forces and momentum	Physics HL/SL A.2 Forces and momentum	Physics HL/SL A.3 Work, energy and power
Two objects of equal mass collide and stick together, one initially at rest. What fraction of the initial kinetic energy is lost? A 0% B 25% C 50% D 100%	A 0.20 kg ball hits a wall at $8.0 \text{ m}\cdot\text{s}^{-1}$ and rebounds at $6.0 \text{ m}\cdot\text{s}^{-1}$ elastically off the wall's normal. Find the impulse delivered to the ball.	State the work-energy theorem and explain what 'net work' means in this context.
IB-PHYSICS-A.2-3 MULTIPLE CHOICE	IB-PHYSICS-A.2-4 PRACTICE	IB-PHYSICS-A.3-1 CONCEPT
Physics HL/SL A.3 Work, energy and power	Physics HL/SL A.3 Work, energy and power	Physics HL/SL A.3 Work, energy and power
Two objects of different mass are dropped from the same height in a vacuum. Do they have the same kinetic energy at the ground? Explain.	A machine does 500 J of useful work while consuming 2000 J of energy input. What is its efficiency? A 10% B 25% C 40% D 75%	A crane lifts a 300 kg load a vertical height of 8.0 m in 12 s at constant velocity. Find the average power delivered.
IB-PHYSICS-A.3-2 CONCEPT	IB-PHYSICS-A.3-3 MULTIPLE CHOICE	IB-PHYSICS-A.3-4 PRACTICE
Physics HL/SL A.4 Rigid body mechanics (HL)	Physics HL/SL A.4 Rigid body mechanics (HL)	Physics HL/SL A.4 Rigid body mechanics (HL)
Why does a solid sphere and a hollow sphere of the same mass and radius roll down an incline at different rates?	State the condition under which angular momentum of a system is conserved.	A wheel has moment of inertia $4.0 \text{ kg}\cdot\text{m}^2$ and angular velocity $10 \text{ rad}\cdot\text{s}^{-1}$. What is its rotational kinetic energy? A 20 J B 40 J C 200 J D 400 J
IB-PHYSICS-A.4-1 CONCEPT	IB-PHYSICS-A.4-2 CONCEPT	IB-PHYSICS-A.4-3 MULTIPLE CHOICE
Physics HL/SL A.4 Rigid body mechanics (HL)	Physics HL/SL A.5 Galilean and special relativity (HL)	Physics HL/SL A.5 Galilean and special relativity (HL)
A torque of $2.0 \text{ N}\cdot\text{m}$ acts on a flywheel with moment of inertia $0.50 \text{ kg}\cdot\text{m}^2$. Find the angular acceleration produced.	State the two postulates of Einstein's special theory of relativity.	Explain why length contraction only affects the dimension along the direction of relative motion.
IB-PHYSICS-A.4-4 PRACTICE	IB-PHYSICS-A.5-1 CONCEPT	IB-PHYSICS-A.5-2 CONCEPT



Physics HL/SL A.5 Galilean and special relativity (HL) As v approaches c , the Lorentz factor γ approaches: A 0 B 1 C infinity D c IB-PHYSICS-A.5-3 MULTIPLE CHOICE	Physics HL/SL A.5 Galilean and special relativity (HL) Find γ for a particle moving at $v = 0.60c$. IB-PHYSICS-A.5-4 PRACTICE	Physics HL/SL B.1 Thermal energy transfers Explain, in terms of particles, why a substance's temperature stays constant during a phase change even while energy is being supplied. IB-PHYSICS-B.1-1 CONCEPT
Physics HL/SL B.1 Thermal energy transfers Distinguish between conduction and convection as mechanisms of thermal energy transfer. IB-PHYSICS-B.1-2 CONCEPT	Physics HL/SL B.1 Thermal energy transfers How much energy is needed to raise the temperature of 2.0 kg of a substance ($c = 900 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$) by 15 K? A 1350 J B 13500 J C 27000 J D 135000 J IB-PHYSICS-B.1-3 MULTIPLE CHOICE	Physics HL/SL B.1 Thermal energy transfers Find the energy released when 0.10 kg of steam at 100°C fully condenses to water at 100°C. ($L_v = 2.26 \times 10^6 \text{ J}\cdot\text{kg}^{-1}$) IB-PHYSICS-B.1-4 PRACTICE
Physics HL/SL B.2 Greenhouse effect Explain why greenhouse gases can let sunlight in but trap heat from escaping. IB-PHYSICS-B.2-1 CONCEPT	Physics HL/SL B.2 Greenhouse effect What is meant by 'albedo' and how does increasing a surface's albedo affect its equilibrium temperature? IB-PHYSICS-B.2-2 CONCEPT	Physics HL/SL B.2 Greenhouse effect According to the Stefan-Boltzmann law, if a black body's absolute temperature doubles, its radiated power per unit area: A Doubles B Quadruples C Increases by a factor of 8 D Increases by a factor of 16 IB-PHYSICS-B.2-3 MULTIPLE CHOICE
Physics HL/SL B.2 Greenhouse effect A star has surface temperature 10000 K. Find its peak emission wavelength using Wien's law (constant = $2.90 \times 10^{-3} \text{ m}\cdot\text{K}$). IB-PHYSICS-B.2-4 PRACTICE	Physics HL/SL B.3 Gas laws State two assumptions of the kinetic model of an ideal gas. IB-PHYSICS-B.3-1 CONCEPT	Physics HL/SL B.3 Gas laws Explain why real gases deviate from ideal gas behaviour at high pressure and low temperature. IB-PHYSICS-B.3-2 CONCEPT
Physics HL/SL B.3 Gas laws A fixed mass of ideal gas at constant temperature has its volume halved. What happens to its pressure? A Halves B Stays the same C Doubles D Quadruples IB-PHYSICS-B.3-3 MULTIPLE CHOICE	Physics HL/SL B.3 Gas laws Find the number of moles of gas in a 0.010 m ³ container at 200 kPa and 350 K ($R = 8.31 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$). IB-PHYSICS-B.3-4 PRACTICE	Physics HL/SL B.4 Thermodynamics (HL) State the first law of thermodynamics and define each term. IB-PHYSICS-B.4-1 CONCEPT
Physics HL/SL B.4 Thermodynamics (HL) Explain why no real heat engine can achieve 100% efficiency, even in principle. IB-PHYSICS-B.4-2 CONCEPT	Physics HL/SL B.4 Thermodynamics (HL) In an adiabatic process on an ideal gas, which quantity is zero? A ΔU B W C Q D ΔT IB-PHYSICS-B.4-3 MULTIPLE CHOICE	Physics HL/SL B.4 Thermodynamics (HL) A gas is compressed adiabatically, and 400 J of work is done on it. Find its change in internal energy. IB-PHYSICS-B.4-4 PRACTICE



Physics HL/SL B.5 Current and circuits	Physics HL/SL B.5 Current and circuits	Physics HL/SL B.5 Current and circuits
State Ohm's law and explain what it means for a conductor to be 'ohmic'.	Explain why the terminal voltage of a battery decreases as more current is drawn from it.	A 12 Ω resistor carries a current of 0.50 A. What power does it dissipate? A 3.0 W B 6.0 W C 12 W D 24 W
IB-PHYSICS-B.5-1 CONCEPT	IB-PHYSICS-B.5-2 CONCEPT	IB-PHYSICS-B.5-3 MULTIPLE CHOICE
Physics HL/SL B.5 Current and circuits	Physics HL/SL C.1 Simple harmonic motion	Physics HL/SL C.1 Simple harmonic motion
Find the resistance of a wire of length 2.0 m, cross-sectional area $1.0 \times 10^{-6} \text{ m}^2$, and resistivity $1.7 \times 10^{-8} \Omega \cdot \text{m}$.	State the defining condition for simple harmonic motion in terms of acceleration and displacement.	Describe how kinetic and potential energy vary during one full cycle of SHM.
IB-PHYSICS-B.5-4 PRACTICE	IB-PHYSICS-C.1-1 CONCEPT	IB-PHYSICS-C.1-2 CONCEPT
Physics HL/SL C.1 Simple harmonic motion	Physics HL/SL C.1 Simple harmonic motion	Physics HL/SL C.2 Wave model
A pendulum's period does NOT depend significantly on which of the following (for small oscillations)? A Length of the pendulum B Gravitational field strength C Amplitude of oscillation D None — it depends on all three	An oscillator has angular frequency $5.0 \text{ rad} \cdot \text{s}^{-1}$ and amplitude 0.040 m. Find its maximum acceleration.	Distinguish between a transverse wave and a longitudinal wave, giving one example of each.
IB-PHYSICS-C.1-3 MULTIPLE CHOICE	IB-PHYSICS-C.1-4 PRACTICE	IB-PHYSICS-C.2-1 CONCEPT
Physics HL/SL C.2 Wave model	Physics HL/SL C.2 Wave model	Physics HL/SL C.2 Wave model
Why does the intensity of a wave from a point source decrease with the square of distance rather than linearly?	Which of the following correctly orders regions of the electromagnetic spectrum by increasing frequency? A Radio, visible, X-ray, gamma B Gamma, X-ray, visible, radio C Visible, radio, gamma, X-ray D X-ray, gamma, radio, visible	A source emits waves of wavelength 0.60 m at a speed of $1500 \text{ m} \cdot \text{s}^{-1}$. Find the frequency.
IB-PHYSICS-C.2-2 CONCEPT	IB-PHYSICS-C.2-3 MULTIPLE CHOICE	IB-PHYSICS-C.2-4 PRACTICE
Physics HL/SL C.3 Wave phenomena	Physics HL/SL C.3 Wave phenomena	Physics HL/SL C.3 Wave phenomena
State the condition for two coherent waves to produce a bright (constructive interference) fringe.	Explain why diffraction effects are much more noticeable for sound waves passing through a doorway than for light waves passing through the same doorway.	Light passes from air ($n=1.0$) into water ($n=1.33$) at an angle of incidence of 40° . What happens to its speed and direction? A Speed increases, bends away from normal B Speed decreases, bends towards normal C Speed decreases, bends away from normal D Speed stays the same, bends towards normal
IB-PHYSICS-C.3-1 CONCEPT	IB-PHYSICS-C.3-2 CONCEPT	IB-PHYSICS-C.3-3 MULTIPLE CHOICE



Physics HL/SL C.3 Wave phenomena	Physics HL/SL C.4 Standing waves and resonance	Physics HL/SL C.4 Standing waves and resonance
Find the critical angle for light travelling from a medium of refractive index 1.50 into air ($n=1.00$).	Explain the key difference between a node and an antinode in a standing wave.	Why can an open-ended pipe support all harmonics but a pipe closed at one end can only support odd harmonics?
IB-PHYSICS-C.3-4PRACTICE	IB-PHYSICS-C.4-1CONCEPT	IB-PHYSICS-C.4-2CONCEPT
Physics HL/SL C.4 Standing waves and resonance	Physics HL/SL C.4 Standing waves and resonance	Physics HL/SL C.5 Doppler effect
A driving force is applied to an oscillating system at a frequency equal to its natural frequency. What happens to the amplitude of oscillation (with low damping)? A It stays at the driving amplitude B It becomes very large (resonance) C It drops to zero D It oscillates chaotically	An open pipe of length 0.68 m resonates at its fundamental with speed of sound $340 \text{ m}\cdot\text{s}^{-1}$. Find the fundamental frequency.	Explain physically why an approaching sound source causes a higher observed frequency.
IB-PHYSICS-C.4-3MULTIPLE CHOICE	IB-PHYSICS-C.4-4PRACTICE	IB-PHYSICS-C.5-1CONCEPT
Physics HL/SL C.5 Doppler effect	Physics HL/SL C.5 Doppler effect	Physics HL/SL C.5 Doppler effect
Why is the Doppler effect used to detect exoplanets and measure the expansion of the universe?	A source of sound moves away from a stationary observer. Compared to the source frequency, the observed frequency is: A Higher B Lower C The same D Cannot be determined	A source emitting 400 Hz moves away from a stationary observer at $20 \text{ m}\cdot\text{s}^{-1}$ (speed of sound = $340 \text{ m}\cdot\text{s}^{-1}$). Find the observed frequency.
IB-PHYSICS-C.5-2CONCEPT	IB-PHYSICS-C.5-3MULTIPLE CHOICE	IB-PHYSICS-C.5-4PRACTICE
Physics HL/SL D.1 Gravitational fields	Physics HL/SL D.1 Gravitational fields	Physics HL/SL D.1 Gravitational fields
Define gravitational field strength and state its SI units.	Explain why gravitational potential energy is defined as negative for a bound orbiting system.	According to Kepler's third law for circular orbits, T^2 is proportional to: A r B r^2 C r^3 D $1/r$
IB-PHYSICS-D.1-1CONCEPT	IB-PHYSICS-D.1-2CONCEPT	IB-PHYSICS-D.1-3MULTIPLE CHOICE
Physics HL/SL D.1 Gravitational fields	Physics HL/SL D.2 Electric and magnetic fields	Physics HL/SL D.2 Electric and magnetic fields
Find the gravitational force between two 1000 kg masses separated by 2.0 m ($G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$).	State Coulomb's law and explain how the force direction depends on the signs of the two charges.	Explain why a charged particle moving parallel to a magnetic field experiences no magnetic force.
IB-PHYSICS-D.1-4PRACTICE	IB-PHYSICS-D.2-1CONCEPT	IB-PHYSICS-D.2-2CONCEPT
Physics HL/SL D.2 Electric and magnetic fields	Physics HL/SL D.2 Electric and magnetic fields	Physics HL/SL D.3 Motion in electromagnetic fields
A point charge creates an electric field. If the distance from the charge is tripled, the field strength becomes: A 3 times as strong B 1/3 as strong C 1/9 as strong D 9 times as strong	A charge of $2.0 \mu\text{C}$ is placed in a uniform electric field of strength $5.0 \times 10^3 \text{ V}\cdot\text{m}^{-1}$. Find the force on the charge.	Explain why a charged particle moving in a uniform magnetic field perpendicular to its velocity travels in a circle rather than speeding up or slowing down.
IB-PHYSICS-D.2-3MULTIPLE CHOICE	IB-PHYSICS-D.2-4PRACTICE	IB-PHYSICS-D.3-1CONCEPT



Physics HL/SL D.3 Motion in electromagnetic fields	Physics HL/SL D.3 Motion in electromagnetic fields	Physics HL/SL D.3 Motion in electromagnetic fields
In a velocity selector, why do particles with the correct speed travel straight through while others are deflected?	A charged particle enters a region of uniform electric field E , moving perpendicular to the field lines with no magnetic field present. What type of path does it follow? A Straight line at constant speed B Circular path C Parabolic path D Helical path	An ion of charge $3.2 \times 10^{-19} \text{ C}$ moves at $1.0 \times 10^5 \text{ m} \cdot \text{s}^{-1}$ perpendicular to a 0.40 T magnetic field, with mass $6.6 \times 10^{-26} \text{ kg}$. Find the radius of its circular path.
IB-PHYSICS-D.3-2 CONCEPT	IB-PHYSICS-D.3-3 MULTIPLE CHOICE	IB-PHYSICS-D.3-4 PRACTICE
Physics HL/SL D.4 Electromagnetic induction (HL)	Physics HL/SL D.4 Electromagnetic induction (HL)	Physics HL/SL D.4 Electromagnetic induction (HL)
State Lenz's law and explain the physical principle it reflects.	Explain why the emf induced in a rotating coil generator varies sinusoidally with time rather than being constant.	An ideal transformer has 100 turns on its primary coil and 500 turns on its secondary coil. If the primary voltage is 20 V , what is the secondary voltage? A 4 V B 20 V C 100 V D 500 V
IB-PHYSICS-D.4-1 CONCEPT	IB-PHYSICS-D.4-2 CONCEPT	IB-PHYSICS-D.4-3 MULTIPLE CHOICE
Physics HL/SL D.4 Electromagnetic induction (HL)	Physics HL/SL E.1 Structure of the atom	Physics HL/SL E.1 Structure of the atom
A coil experiences a flux change of 0.020 Wb in 0.10 s across its 50 turns. Find the magnitude of the induced emf.	Explain why atomic emission spectra consist of discrete lines rather than a continuous range of wavelengths.	What is the key difference between isotopes of the same element?
IB-PHYSICS-D.4-4 PRACTICE	IB-PHYSICS-E.1-1 CONCEPT	IB-PHYSICS-E.1-2 CONCEPT
Physics HL/SL E.1 Structure of the atom	Physics HL/SL E.1 Structure of the atom	Physics HL/SL E.2 Quantum physics (HL)
An atom labelled ${}^{14}_6\text{C}$ has how many neutrons? A 6 B 8 C 14 D 20	Find the energy in joules of a photon with frequency $5.0 \times 10^{14} \text{ Hz}$ ($h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$).	Explain why the photoelectric effect could not be explained by the classical wave theory of light, but was explained by Einstein's photon model.
IB-PHYSICS-E.1-3 MULTIPLE CHOICE	IB-PHYSICS-E.1-4 PRACTICE	IB-PHYSICS-E.2-1 CONCEPT
Physics HL/SL E.2 Quantum physics (HL)	Physics HL/SL E.2 Quantum physics (HL)	Physics HL/SL E.2 Quantum physics (HL)
State the de Broglie hypothesis and give the formula for the de Broglie wavelength.	According to the Heisenberg uncertainty principle, if the position of a particle is measured with very high precision (Δx very small), what happens to the minimum uncertainty in its momentum? A It becomes very small too B It becomes very large C It stays the same D It becomes exactly zero	Find the de Broglie wavelength of a 0.20 kg ball moving at $15 \text{ m} \cdot \text{s}^{-1}$ ($h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$).
IB-PHYSICS-E.2-2 CONCEPT	IB-PHYSICS-E.2-3 MULTIPLE CHOICE	IB-PHYSICS-E.2-4 PRACTICE



Physics HL/SL E.3 Radioactive decay	Physics HL/SL E.3 Radioactive decay	Physics HL/SL E.3 Radioactive decay
Explain why radioactive decay is described as a random process, and how this is reconciled with the predictable exponential decay of a large sample.	State what is conserved in all types of radioactive decay (alpha, beta, gamma).	A sample has a half-life of 10 years. What fraction of the original undecayed nuclei remains after 30 years? A 1/2 B 1/4 C 1/8 D 1/16
IB-PHYSICS-E.3-1 CONCEPT	IB-PHYSICS-E.3-2 CONCEPT	IB-PHYSICS-E.3-3 MULTIPLE CHOICE
Physics HL/SL E.3 Radioactive decay	Physics HL/SL E.4 Fission	Physics HL/SL E.4 Fission
A source has a decay constant of 0.10 s^{-1} . Find its half-life.	Explain, in terms of binding energy per nucleon, why nuclear fission of a heavy nucleus releases energy.	What is a chain reaction in the context of nuclear fission, and what condition must be met for it to be self-sustaining?
IB-PHYSICS-E.3-4 PRACTICE	IB-PHYSICS-E.4-1 CONCEPT	IB-PHYSICS-E.4-2 CONCEPT
Physics HL/SL E.4 Fission	Physics HL/SL E.4 Fission	Physics HL/SL E.5 Fusion and stars
What is the role of control rods in a nuclear fission reactor? A To speed up neutrons B To absorb neutrons and regulate the reaction rate C To act as the fuel source D To convert kinetic energy directly into electricity	Find the energy equivalent of a mass defect of $1.0 \times 10^{-29}\text{ kg}$ ($c = 3.00 \times 10^8\text{ m}\cdot\text{s}^{-1}$).	Explain why fusing light nuclei releases energy, referencing binding energy per nucleon.
IB-PHYSICS-E.4-3 MULTIPLE CHOICE	IB-PHYSICS-E.4-4 PRACTICE	IB-PHYSICS-E.5-1 CONCEPT
Physics HL/SL E.5 Fusion and stars	Physics HL/SL E.5 Fusion and stars	Physics HL/SL E.5 Fusion and stars
What keeps a main-sequence star in stable equilibrium, neither collapsing nor expanding indefinitely?	On a Hertzsprung-Russell (HR) diagram, where are the hottest, most luminous stars generally located? A Top left B Top right C Bottom left D Bottom right	Find the energy released when a mass defect of $2.0 \times 10^{-28}\text{ kg}$ is converted to energy in a fusion reaction ($c = 3.00 \times 10^8\text{ m}\cdot\text{s}^{-1}$).
IB-PHYSICS-E.5-2 CONCEPT	IB-PHYSICS-E.5-3 MULTIPLE CHOICE	IB-PHYSICS-E.5-4 PRACTICE



Physics HL/SL A.1 Kinematics	Physics HL/SL A.1 Kinematics	Physics HL/SL A.1 Kinematics
C. $s = \frac{1}{2}gt^2 = \frac{1}{2}(9.81)(9.0) = 44.1$ m.	Still g ($\approx 9.81 \text{ m}\cdot\text{s}^{-2}$, downward) — velocity is zero at the top, but acceleration is never zero since gravity acts throughout the flight.	Because gravity acts only vertically, giving perpendicular components of velocity that do not affect each other; horizontal velocity stays constant while vertical velocity changes uniformly under g .
IB-PHYSICS-A.1-3 ANSWER	IB-PHYSICS-A.1-2 ANSWER	IB-PHYSICS-A.1-1 ANSWER
Physics HL/SL A.2 Forces and momentum	Physics HL/SL A.2 Forces and momentum	Physics HL/SL A.1 Kinematics
Momentum is conserved because the net external force on an isolated system is zero (internal forces cancel in pairs by Newton's third law); kinetic energy can convert into heat, sound or deformation energy, so it is only conserved if the collision is perfectly elastic.	For every action force one body exerts on a second, the second exerts an equal and opposite force on the first; they act on different objects, so they cannot cancel—only forces on the same object can be summed to give a net force.	$v^2 = u^2 + 2as \rightarrow 0 = 64 + 2a(20) \rightarrow a = -1.6 \text{ m}\cdot\text{s}^{-2}$. Deceleration = $1.6 \text{ m}\cdot\text{s}^{-2}$.
IB-PHYSICS-A.2-2 ANSWER	IB-PHYSICS-A.2-1 ANSWER	IB-PHYSICS-A.1-4 ANSWER
Physics HL/SL A.3 Work, energy and power	Physics HL/SL A.2 Forces and momentum	Physics HL/SL A.2 Forces and momentum
The net work done by all forces acting on an object equals its change in kinetic energy, $\Delta E_k = W_{\text{net}}$; 'net' means the vector sum of all forces' work, not any single force acting alone.	Taking incoming direction as positive: $\Delta p = m(v_f - v_i) = 0.20(-6.0 - 8.0) = -2.8 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$. Impulse magnitude = $2.8 \text{ N}\cdot\text{s}$, directed away from the wall.	C. For equal masses, one moving at v and one at rest sticking together: final speed = $v/2$; $KE_f/KE_i = \frac{1}{4}$, so 50% of KE is lost.
IB-PHYSICS-A.3-1 ANSWER	IB-PHYSICS-A.2-4 ANSWER	IB-PHYSICS-A.2-3 ANSWER
Physics HL/SL A.3 Work, energy and power	Physics HL/SL A.3 Work, energy and power	Physics HL/SL A.3 Work, energy and power
$W = mgh = 300 \times 9.81 \times 8.0 = 23544 \text{ J}$. $P = W/t = 23544/12 = 1962$ $W \approx 2.0 \times 10^3 \text{ W}$.	B. Efficiency = useful output/input = $500/2000 = 0.25 = 25\%$.	No — both fall with the same acceleration and reach the same final speed, but $E_k = \frac{1}{2}mv^2$ depends on mass, so the heavier object has more kinetic energy at the ground even though speeds are equal.
IB-PHYSICS-A.3-4 ANSWER	IB-PHYSICS-A.3-3 ANSWER	IB-PHYSICS-A.3-2 ANSWER
Physics HL/SL A.4 Rigid body mechanics (HL)	Physics HL/SL A.4 Rigid body mechanics (HL)	Physics HL/SL A.4 Rigid body mechanics (HL)
C. $E_k = \frac{1}{2}I\omega^2 = \frac{1}{2}(4.0)(100) = 200$ J.	Angular momentum is conserved when the net external torque acting on the system is zero.	They have different moments of inertia (solid sphere $I = \frac{2}{5}MR^2$, hollow sphere $I = \frac{2}{3}MR^2$); more of the hollow sphere's mass is farther from the axis, so it has greater rotational inertia, gaining less angular acceleration for the same torque and rolling down more slowly.
IB-PHYSICS-A.4-3 ANSWER	IB-PHYSICS-A.4-2 ANSWER	IB-PHYSICS-A.4-1 ANSWER



Physics HL/SL A.5 Galilean and special relativity (HL)	Physics HL/SL A.5 Galilean and special relativity (HL)	Physics HL/SL A.4 Rigid body mechanics (HL)
Length contraction arises from the relativity of simultaneity affecting measurements made along the direction of travel; dimensions perpendicular to the motion involve no such simultaneity discrepancy, so they are unaffected.	1) The laws of physics are the same in all inertial reference frames. 2) The speed of light in a vacuum, c , has the same value for all observers, independent of the motion of the source or the observer.	$\alpha = \tau/I = 2.0/0.50 = 4.0 \text{ rad}\cdot\text{s}^{-2}$.
IB-PHYSICS-A.5-2ANSWER	IB-PHYSICS-A.5-1ANSWER	IB-PHYSICS-A.4-4ANSWER
Physics HL/SL B.1 Thermal energy transfers	Physics HL/SL A.5 Galilean and special relativity (HL)	Physics HL/SL A.5 Galilean and special relativity (HL)
The added energy goes into breaking/weakening intermolecular bonds (increasing potential energy of the particles) rather than increasing their average kinetic energy, so temperature (which depends on kinetic energy) does not change during the phase change.	$\gamma = 1/\sqrt{1-0.36} = 1/\sqrt{0.64} = 1/0.80 = 1.25$.	C. $\gamma = 1/\sqrt{1-v^2/c^2}$; as $v \rightarrow c$, the denominator $\rightarrow 0$, so $\gamma \rightarrow \text{infinity}$.
IB-PHYSICS-B.1-1ANSWER	IB-PHYSICS-A.5-4ANSWER	IB-PHYSICS-A.5-3ANSWER
Physics HL/SL B.1 Thermal energy transfers	Physics HL/SL B.1 Thermal energy transfers	Physics HL/SL B.1 Thermal energy transfers
$Q = mL = 0.10 \times 2.26 \times 10^6 = 2.26 \times 10^5 \text{ J}$ released.	C. $Q = mc\Delta T = 2.0 \times 900 \times 15 = 27000 \text{ J}$.	Conduction transfers energy through particle-to-particle collisions/vibrations without bulk movement of the medium (dominant in solids); convection transfers energy through bulk movement of a fluid, where warmer, less dense fluid rises and is replaced by cooler, denser fluid.
IB-PHYSICS-B.1-4ANSWER	IB-PHYSICS-B.1-3ANSWER	IB-PHYSICS-B.1-2ANSWER
Physics HL/SL B.2 Greenhouse effect	Physics HL/SL B.2 Greenhouse effect	Physics HL/SL B.2 Greenhouse effect
D. Power per unit area $\propto T^4$; doubling T increases it by $2^4 = 16$.	Albedo is the fraction of incident electromagnetic radiation reflected by a surface rather than absorbed; a higher albedo means less solar energy is absorbed, tending to lower the equilibrium temperature of that surface.	Greenhouse gas molecules have vibrational/rotational energy levels that strongly absorb longwave infrared radiation (matching Earth's emission spectrum) but are largely transparent to the shorter visible/UV wavelengths that dominate incoming solar radiation, so incoming energy passes through while outgoing infrared is absorbed and partly re-radiated back to the surface.
IB-PHYSICS-B.2-3ANSWER	IB-PHYSICS-B.2-2ANSWER	IB-PHYSICS-B.2-1ANSWER



Physics HL/SL B.3 Gas laws At high pressure particle volume becomes non-negligible compared to the container volume, and at low temperature intermolecular attractive forces (negligible in the ideal model) become significant relative to the particles' reduced kinetic energy, both causing deviation from $pV=nRT$. IB-PHYSICS-B.3-2 ANSWER	Physics HL/SL B.3 Gas laws Any two of: gas particles are point-like (negligible volume compared to container); collisions between particles and with walls are perfectly elastic; there are no intermolecular forces except during collisions; particles move in continuous random motion; the time of collision is negligible compared to time between collisions. IB-PHYSICS-B.3-1 ANSWER	Physics HL/SL B.2 Greenhouse effect $\lambda_{\text{max}} = 2.90 \times 10^{-3} / 10000 = 2.90 \times 10^{-7} \text{ m}$ (in the ultraviolet). IB-PHYSICS-B.2-4 ANSWER
Physics HL/SL B.4 Thermodynamics (HL) $\Delta U = Q + W$, where ΔU is the change in internal energy of the system, Q is the thermal energy (heat) transferred to the system, and W is the work done on the system by its surroundings. IB-PHYSICS-B.4-1 ANSWER	Physics HL/SL B.3 Gas laws $n = pV/RT = (2.0 \times 10^5)(0.010)/(8.31 \times 350) = 2000/2908.5 = 0.688 \text{ mol}$. IB-PHYSICS-B.3-4 ANSWER	Physics HL/SL B.3 Gas laws C. At constant T, $pV = \text{constant}$ (Boyle's law), so halving V doubles p. IB-PHYSICS-B.3-3 ANSWER
Physics HL/SL B.4 Thermodynamics (HL) $Q = 0$ (adiabatic), so $\Delta U = Q + W = 0 + 400 = 400 \text{ J}$ increase. IB-PHYSICS-B.4-4 ANSWER	Physics HL/SL B.4 Thermodynamics (HL) C. Adiabatic means no heat is exchanged with the surroundings, so $Q = 0$. IB-PHYSICS-B.4-3 ANSWER	Physics HL/SL B.4 Thermodynamics (HL) The second law of thermodynamics requires that some heat must always be rejected to a cold reservoir in any cyclic heat engine (entropy of an isolated system cannot decrease); even an ideal reversible (Carnot) engine has efficiency $1 - T_c/T_h < 1$ unless $T_c = 0 \text{ K}$, which is unattainable. IB-PHYSICS-B.4-2 ANSWER
Physics HL/SL B.5 Current and circuits A. $P = I^2 R = (0.50)^2(12) = 3.0 \text{ W}$. IB-PHYSICS-B.5-3 ANSWER	Physics HL/SL B.5 Current and circuits A real battery has internal resistance r; some of the emf is used to drive current through this internal resistance ($V_{\text{internal}} = Ir$), so the terminal voltage $V = \mathcal{E} - Ir$ decreases as the current I increases. IB-PHYSICS-B.5-2 ANSWER	Physics HL/SL B.5 Current and circuits Ohm's law states $V = IR$ where R is constant for a given conductor at constant temperature; a conductor is ohmic if its resistance (the V/I ratio) remains constant as the current (and hence voltage) through it varies, giving a straight-line V against I graph through the origin. IB-PHYSICS-B.5-1 ANSWER
Physics HL/SL C.1 Simple harmonic motion Kinetic energy is maximum at equilibrium ($x=0$, where speed is maximum) and zero at maximum displacement; potential energy is zero at equilibrium and maximum at maximum displacement; their sum (total mechanical energy) remains constant throughout, continuously converting from one form to the other. IB-PHYSICS-C.1-2 ANSWER	Physics HL/SL C.1 Simple harmonic motion The acceleration must be directly proportional to the displacement from a fixed equilibrium point and always directed towards that point (opposite in sign to the displacement): $a = -\omega^2 x$. IB-PHYSICS-C.1-1 ANSWER	Physics HL/SL B.5 Current and circuits $R = \rho L/A = (1.7 \times 10^{-8})(2.0)/(1.0 \times 10^{-6}) = 3.4 \times 10^{-2} \Omega$. IB-PHYSICS-B.5-4 ANSWER



Physics HL/SL C.2 Wave model	Physics HL/SL C.1 Simple harmonic motion	Physics HL/SL C.1 Simple harmonic motion
In a transverse wave, particle oscillation is perpendicular to the direction of wave travel/energy transfer (e.g. light or a wave on a string); in a longitudinal wave, oscillation is parallel to the direction of travel (e.g. sound in air).	$a_{\text{max}} = \omega^2 x_0 = (5.0)^2(0.040) = 1.0 \text{ m}\cdot\text{s}^{-2}.$	C. For small-angle oscillations, the period $T = 2\pi\sqrt{l/g}$ is independent of amplitude.
IB-PHYSICS-C.2-1 ANSWER	IB-PHYSICS-C.1-4 ANSWER	IB-PHYSICS-C.1-3 ANSWER
Physics HL/SL C.2 Wave model	Physics HL/SL C.2 Wave model	Physics HL/SL C.2 Wave model
$f = v/\lambda = 1500/0.60 = 2500 \text{ Hz}.$	A. Frequency increases from radio (lowest) through visible light to X-ray and gamma rays (highest).	The wave's power spreads out over the surface of an expanding sphere of area $4\pi r^2$; since intensity is power divided by area, and area grows as r^2 , intensity falls as $1/r^2$ (assuming no absorption).
IB-PHYSICS-C.2-4 ANSWER	IB-PHYSICS-C.2-3 ANSWER	IB-PHYSICS-C.2-2 ANSWER
Physics HL/SL C.3 Wave phenomena	Physics HL/SL C.3 Wave phenomena	Physics HL/SL C.3 Wave phenomena
B. Water has a higher refractive index, so light slows down and bends towards the normal (Snell's law).	Diffraction is significant when the aperture size is comparable to the wavelength; sound wavelengths (order of cm to m) are comparable to typical doorway widths, producing noticeable spreading, whereas visible light wavelengths ($\sim 500 \text{ nm}$) are far smaller than the doorway, so diffraction of light there is negligible.	The path difference between the two waves at that point must be a whole number of wavelengths: path difference $= m\lambda$, where $m = 0, \pm 1, \pm 2, \dots$
IB-PHYSICS-C.3-3 ANSWER	IB-PHYSICS-C.3-2 ANSWER	IB-PHYSICS-C.3-1 ANSWER
Physics HL/SL C.4 Standing waves and resonance	Physics HL/SL C.4 Standing waves and resonance	Physics HL/SL C.3 Wave phenomena
A closed end must be a displacement node and an open end must be an antinode; fitting this boundary condition (one node, one antinode) into the pipe length only allows odd multiples of the fundamental, whereas an open-open pipe (antinode at both ends) allows both even and odd harmonics.	A node is a point of permanent zero (or minimum) displacement where the two superposing waves are always in antiphase and cancel; an antinode is a point of maximum displacement amplitude where the two waves are always in phase and reinforce each other.	$\sin\theta_c = n_2/n_1 = 1.00/1.50 = 0.667$ $\rightarrow \theta_c = 41.8^\circ.$
IB-PHYSICS-C.4-2 ANSWER	IB-PHYSICS-C.4-1 ANSWER	IB-PHYSICS-C.3-4 ANSWER
Physics HL/SL C.5 Doppler effect	Physics HL/SL C.4 Standing waves and resonance	Physics HL/SL C.4 Standing waves and resonance
As the source moves towards the observer, each successive wavefront is emitted from a position closer to the observer than the previous one, compressing the wavelength in that direction; since wave speed in the medium is unchanged, a shorter wavelength corresponds to a higher observed frequency ($f=v/\lambda$).	$f_1 = v/(2L) = 340/(2\times 0.68) = 340/1.36 = 250 \text{ Hz}.$	B. When driving frequency matches natural frequency, resonance occurs and the amplitude of oscillation becomes very large, limited only by damping.
IB-PHYSICS-C.5-1 ANSWER	IB-PHYSICS-C.4-4 ANSWER	IB-PHYSICS-C.4-3 ANSWER



Physics HL/SL C.5 Doppler effect $f_{\text{obs}} = f(v)/(v+u_s) = 400(340)/(340+20) = 136000/360 = 378 \text{ Hz}.$ IB-PHYSICS-C.5-4 ANSWER	Physics HL/SL C.5 Doppler effect B. A receding source stretches the wavefronts behind it, producing a lower observed frequency than the source frequency. IB-PHYSICS-C.5-3 ANSWER	Physics HL/SL C.5 Doppler effect The periodic Doppler shift in a star's spectral lines reveals a wobble caused by an orbiting planet's gravitational pull (radial velocity method for exoplanets), while systematic redshift of distant galaxies' spectra indicates they are receding, providing evidence for the expansion of the universe. IB-PHYSICS-C.5-2 ANSWER
Physics HL/SL D.1 Gravitational fields C. From $GMm/r^2 = m\omega^2 r$ and $T=2\pi/\omega$, one derives $T^2 = (4\pi^2/GM)r^3$, so $T^2 \propto r^3$. IB-PHYSICS-D.1-3 ANSWER	Physics HL/SL D.1 Gravitational fields Gravitational potential energy is taken as zero at infinite separation (where there is no interaction); since gravity is attractive, work must be done ON the system to increase separation (move it towards zero), meaning energy at any finite separation is lower than at infinity, hence negative. IB-PHYSICS-D.1-2 ANSWER	Physics HL/SL D.1 Gravitational fields Gravitational field strength g at a point is the gravitational force per unit mass experienced by a small test mass placed at that point, $g = F/m$, measured in $\text{N}\cdot\text{kg}^{-1}$ (equivalent to $\text{m}\cdot\text{s}^{-2}$). IB-PHYSICS-D.1-1 ANSWER
Physics HL/SL D.2 Electric and magnetic fields The magnetic force magnitude is $F = qvB\sin\theta$, where θ is the angle between v and B; when the particle moves parallel (or antiparallel) to B, $\theta = 0^\circ$ (or 180°), so $\sin\theta = 0$ and the force is zero. IB-PHYSICS-D.2-2 ANSWER	Physics HL/SL D.2 Electric and magnetic fields $F = kq_1q_2/r^2$, acting along the line joining the two point charges; if the charges have the same sign the force is repulsive (pushing them apart), and if they have opposite signs the force is attractive (pulling them together). IB-PHYSICS-D.2-1 ANSWER	Physics HL/SL D.1 Gravitational fields $F = Gm_1m_2/r^2 = (6.67\times 10^{-11})(1000)(1000)/(2.0)^2 = 6.67\times 10^{-5}/4.0 = 1.67\times 10^{-5} \text{ N}.$ IB-PHYSICS-D.1-4 ANSWER
Physics HL/SL D.3 Motion in electromagnetic fields The magnetic force is always perpendicular to the particle's velocity ($F=qv\times B$), so it changes the direction of motion but does no work on the particle (force and displacement are perpendicular) and cannot change its speed; a constant-magnitude force always perpendicular to velocity produces uniform circular motion. IB-PHYSICS-D.3-1 ANSWER	Physics HL/SL D.2 Electric and magnetic fields $F = qE = (2.0\times 10^{-6})(5.0\times 10^3) = 1.0\times 10^{-2} \text{ N}.$ IB-PHYSICS-D.2-4 ANSWER	Physics HL/SL D.2 Electric and magnetic fields C. $E = kQ/r^2 \propto 1/r^2$; tripling r reduces E to 1/9 of its original value. IB-PHYSICS-D.2-3 ANSWER
Physics HL/SL D.3 Motion in electromagnetic fields $r = mv/(qB) = (6.6\times 10^{-26})(1.0\times 10^5)/[(3.2\times 10^{-19})(0.40)] = 6.6\times 10^{-21}/1.28\times 10^{-19} = 0.0516 \text{ m}.$ IB-PHYSICS-D.3-4 ANSWER	Physics HL/SL D.3 Motion in electromagnetic fields C. This is analogous to projectile motion: constant velocity perpendicular to E and constant acceleration (qE/m) parallel to E produce a parabolic trajectory. IB-PHYSICS-D.3-3 ANSWER	Physics HL/SL D.3 Motion in electromagnetic fields The electric and magnetic forces are equal and opposite only for particles at the specific speed $v=E/B$, giving zero net force and straight-line motion; particles with other speeds experience an unbalanced net force (electric or magnetic force dominating) and are deflected off the straight path. IB-PHYSICS-D.3-2 ANSWER



Physics HL/SL D.4 Electromagnetic induction (HL) C. $V_s/V_p = N_s/N_p \rightarrow V_s = 20 \times (500/100) = 100 \text{ V}$. IB-PHYSICS-D.4-3 ANSWER	Physics HL/SL D.4 Electromagnetic induction (HL) The emf depends on the rate of change of flux linkage, which is proportional to the component of the coil's area perpendicular to B; as the coil rotates at constant ω, this perpendicular component (and hence the rate of change of flux) varies sinusoidally, giving $\mathcal{E} = \mathcal{E}_0 \sin(\omega t)$. IB-PHYSICS-D.4-2 ANSWER	Physics HL/SL D.4 Electromagnetic induction (HL) Lenz's law states that the direction of an induced emf/current is always such as to oppose the change in magnetic flux that produced it; this reflects conservation of energy, since work must be done against the induced effect to change the flux, and this work is the source of the electrical energy generated. IB-PHYSICS-D.4-1 ANSWER
Physics HL/SL E.1 Structure of the atom Isotopes of an element have the same number of protons (atomic number Z, and hence the same chemical properties) but different numbers of neutrons, giving them different mass numbers A and different nuclear stability/mass. IB-PHYSICS-E.1-2 ANSWER	Physics HL/SL E.1 Structure of the atom Electrons in an atom can only occupy specific quantised energy levels; a photon is emitted only when an electron transitions between two allowed levels, so only photons with energies exactly matching the discrete energy gaps between levels are emitted, producing discrete spectral lines. IB-PHYSICS-E.1-1 ANSWER	Physics HL/SL D.4 Electromagnetic induction (HL) $\mathcal{E} = N(\Delta\Phi/\Delta t) = 50 \times (0.020/0.10) = 50 \times 0.20 = 10 \text{ V}$. IB-PHYSICS-D.4-4 ANSWER
Physics HL/SL E.2 Quantum physics (HL) Classical wave theory predicts that increasing light intensity should increase the energy delivered to electrons regardless of frequency, with emission possible at any frequency given enough time/intensity; experimentally, emission only occurs above a threshold frequency (independent of intensity) with instantaneous emission, which Einstein explained by treating light as photons of discrete energy hf, each interacting with one electron, requiring $hf \geq \phi$ for emission. IB-PHYSICS-E.2-1 ANSWER	Physics HL/SL E.1 Structure of the atom $E = hf = (6.63 \times 10^{-34})(5.0 \times 10^{14}) = 3.315 \times 10^{-19} \text{ J}$. IB-PHYSICS-E.1-4 ANSWER	Physics HL/SL E.1 Structure of the atom B. Mass number $A=14$, atomic number $Z=6$, so neutrons $= A-Z = 14-6 = 8$. IB-PHYSICS-E.1-3 ANSWER
Physics HL/SL E.2 Quantum physics (HL) $\lambda = h/(mv) = 6.63 \times 10^{-34}/(0.20 \times 15) = 6.63 \times 10^{-34}/3.0 = 2.21 \times 10^{-34} \text{ m}$ (immeasurably small, showing why macroscopic objects show no observable wave behaviour). IB-PHYSICS-E.2-4 ANSWER	Physics HL/SL E.2 Quantum physics (HL) B. Since $\Delta x \cdot \Delta p \geq h/4\pi$ (a fixed minimum product), a very small Δx forces Δp to be correspondingly large. IB-PHYSICS-E.2-3 ANSWER	Physics HL/SL E.2 Quantum physics (HL) Every moving particle has an associated wavelength given by $\lambda = h/p = h/(mv)$, where h is Planck's constant and p is the particle's momentum; this reflects wave-particle duality for matter, confirmed experimentally by electron diffraction. IB-PHYSICS-E.2-2 ANSWER



Physics HL/SL E.3 Radioactive decay	Physics HL/SL E.3 Radioactive decay	Physics HL/SL E.3 Radioactive decay
C. 30 years = 3 half-lives, so remaining fraction = $(1/2)^3 = 1/8$.	In every decay, total mass/energy (via $E=mc^2$, including any kinetic energy and rest mass of emitted particles), total charge, mass number A, and atomic number Z (accounting for the charge of the emitted particle in beta decay) are all conserved.	Radioactive decay is random at the level of an individual nucleus — it is impossible to predict exactly when any specific nucleus will decay, and each nucleus has a constant probability of decay per unit time (decay constant λ) regardless of its age; for a very large number of nuclei, this constant probability applied statistically to huge numbers produces a smooth, predictable exponential decrease in the overall population, $N=N_0e^{(-\lambda t)}$.
IB-PHYSICS-E.3-3 ANSWER	IB-PHYSICS-E.3-2 ANSWER	IB-PHYSICS-E.3-1 ANSWER
Physics HL/SL E.4 Fission	Physics HL/SL E.4 Fission	Physics HL/SL E.3 Radioactive decay
A chain reaction occurs when neutrons released from one fission event go on to induce further fission events in other nuclei; it is self-sustaining (critical) when, on average, exactly one neutron per fission causes another fission, keeping the reaction rate constant rather than dying out or growing uncontrollably.	The daughter nuclei produced in fission have a higher average binding energy per nucleon than the original heavy parent nucleus (since binding energy per nucleon peaks near iron/nickel and heavy nuclei like uranium are below this peak); the more tightly bound products correspond to a lower total mass, and this mass difference is released as energy according to $E=\Delta mc^2$.	$T_{1/2} = \ln 2 / \lambda = 0.693 / 0.10 = 6.93 \text{ s}$.
IB-PHYSICS-E.4-2 ANSWER	IB-PHYSICS-E.4-1 ANSWER	IB-PHYSICS-E.3-4 ANSWER
Physics HL/SL E.5 Fusion and stars	Physics HL/SL E.4 Fission	Physics HL/SL E.4 Fission
Light nuclei (below iron/nickel on the binding energy curve) have lower binding energy per nucleon than the heavier product nucleus formed by fusion; since the product is more tightly bound, its total mass is less than the sum of the reactant masses, and this mass defect is released as energy via $E=\Delta mc^2$.	$E = \Delta mc^2 = (1.0 \times 10^{-29})(9.00 \times 10^{16}) = 9.0 \times 10^{-13} \text{ J}$.	B. Control rods absorb a controllable fraction of neutrons, allowing the reactor's power output to be regulated and kept at a stable, critical level.
IB-PHYSICS-E.5-1 ANSWER	IB-PHYSICS-E.4-4 ANSWER	IB-PHYSICS-E.4-3 ANSWER
Physics HL/SL E.5 Fusion and stars	Physics HL/SL E.5 Fusion and stars	Physics HL/SL E.5 Fusion and stars
$E = \Delta mc^2 = (2.0 \times 10^{-28})(9.00 \times 10^{16}) = 1.8 \times 10^{-11} \text{ J}$.	A. The HR diagram plots luminosity (increasing upward) against surface temperature (decreasing to the right), so the hottest, most luminous stars appear in the top left.	Hydrostatic equilibrium: the inward force of gravity is balanced by the outward thermal and radiation pressure produced by nuclear fusion reactions occurring in the star's core.
IB-PHYSICS-E.5-4 ANSWER	IB-PHYSICS-E.5-3 ANSWER	IB-PHYSICS-E.5-2 ANSWER